

Interview Prep Platform — Requirements Specification

AI-Powered Interview Practice with Intelligent Scoring & Feedback | Freeman Yiu · Youcheng Taing

Project: Interview Prep Platform — A rubric-based, AI-powered web app for structured interview practice.
Stack: Next.js · FastAPI · PostgreSQL · sentence-transformers · Docker · Render

SECTION 1 — FUNCTIONAL REQUIREMENTS

Functional requirements describe what the system must do — its features, behaviors, and data operations.

FR-01

User Registration & Authentication — The system shall allow users to register with an email and password, log in securely, and maintain authenticated sessions using JWT tokens.

FR-02

Role Selection — The system shall present users with selectable job role tracks (SWE, Data Science, PM, Behavioral) and load a curated question bank specific to the chosen role upon selection.

FR-03

Interview Question Delivery — The system shall retrieve and display interview questions one at a time from the role-specific question bank, enabling users to practice in a structured session format.

FR-04

Answer Submission — The system shall accept free-text answers from users for each displayed interview question and persist each submission to the database with a timestamp and user reference.

FR-05

AI-Powered Answer Scoring — The system shall embed submitted answers using sentence-transformers (all-MiniLM-L6-v2) and compute cosine similarity scores against predefined rubric components for each question.

FR-06

Multi-Dimensional Score Display — The system shall calculate and display four independent dimension scores (Technical Depth, Clarity, Completeness, Structure) on a scale of 0–100 for every submitted answer.

FR-07

Keyword Coverage Extraction — The system shall identify which key rubric concepts were addressed and which were missing from a submitted answer, and present this as itemized feedback.

FR-08

Improvement Suggestions — The system shall generate actionable improvement suggestions by surfacing missing rubric elements and explaining what a strong answer would have included.

FR-09

Answer History & Progress Tracking — The system shall store all past answers and scores per user and display a chronological history with per-question score trends over time.

FR-10

Radar Chart Visualization — The system shall render an interactive radar chart showing a user's multi-dimensional score profile, updating dynamically as new answers are submitted.

FR-11

Question Bank Management — The system shall support the ability to add, update, and manage role-specific questions and their associated rubric components via an administrative interface or data layer.

FR-12

Embedding Cache — The system shall cache computed embeddings using Redis to avoid redundant computation for repeated rubric or question lookups, reducing AI scoring latency.

Interview Prep Platform — Non-Functional Requirements

Each requirement includes a verifiable, objectively testable acceptance criterion

SECTION 2 — NON-FUNCTIONAL REQUIREMENTS (Performance · Usability · Reliability)

Non-functional requirements define quality attributes. Each entry includes a measurable verification criterion.

NFR-01 Performance

AI scoring response time — The system shall return multi-dimensional scores and feedback to the user within 3 seconds of answer submission under normal load (up to 50 concurrent users).

VERIFICATION CRITERIA:

VERIFICATION: Load test with 50 simulated concurrent users submitting answers. Measure p95 response time from POST /submit

NFR-02 Performance

Embedding cache hit improvement — The system shall reduce redundant embedding computation time by at least 70% for repeated rubric lookups when Redis cache is active.

VERIFICATION CRITERIA:

VERIFICATION: Run 100 identical rubric embedding requests with cache disabled (baseline), then with cache enabled. Pass if

NFR-03 Performance

Page load time — All primary UI pages (role selection, Q&A interface, dashboard) shall achieve a Largest Contentful Paint (LCP) of under 2.5 seconds on a standard broadband connection.

VERIFICATION CRITERIA:

VERIFICATION: Run Lighthouse audit on each primary page 3 times from a standard test environment. Pass if $LCP \leq 2,500$ ms in all

NFR-04 Usability

Task completion without assistance — A first-time user shall be able to select a role, answer a question, and view their score without any external guidance or documentation.

VERIFICATION CRITERIA:

VERIFICATION: Conduct usability test with 5 new users. Observe and record task completion. Pass if ≥ 4 out of 5 users complete all

NFR-05 Usability

Mobile responsiveness — The platform shall be fully functional and readable on screen widths from 375px (iPhone SE) to 1440px (desktop), with no horizontal scrolling or broken layouts.

VERIFICATION CRITERIA:

VERIFICATION: Test all pages in Chrome DevTools at 375px, 768px, and 1440px widths. Pass if no horizontal scroll appears and all

NFR-06 Reliability

System uptime — The deployed platform on Render shall maintain $\geq 99\%$ uptime during the 9-week evaluation period, excluding scheduled maintenance windows.

VERIFICATION CRITERIA:

VERIFICATION: Monitor uptime using an external service (e.g., UptimeRobot) for the full 9-week period. Pass if total downtime ≤ 1.08 hours (1% of 108 hours).

Interview Prep Platform — Non-Functional Requirements (continued)

Security · Maintainability · Scalability · Data Integrity

SECTION 2 — NON-FUNCTIONAL REQUIREMENTS (Security · Maintainability · Scalability · Data Integrity)

NFR-07

Security

Authentication security — All protected API endpoints shall reject requests missing a valid JWT token with a 401 Unauthorized response. Tokens shall expire after 24 hours.

VERIFICATION CRITERIA:

VERIFICATION: Send 20 API requests to protected endpoints without a token, with an expired token, and with a valid token. Pass if

NFR-08

Security

Password storage — User passwords shall never be stored in plaintext. All passwords shall be hashed using bcrypt with a minimum cost factor of 12 before database persistence.

VERIFICATION CRITERIA:

VERIFICATION: Register a test user, then inspect the stored password hash in the database. Pass if the stored value is a bcrypt

NFR-09

Maintainability

CI pipeline — All code merged to the main branch shall automatically pass a GitHub Actions CI pipeline that runs unit tests, lint checks, and a Docker build within 5 minutes.

VERIFICATION CRITERIA:

VERIFICATION: Merge a pull request and observe the GitHub Actions run. Pass if the pipeline completes in ≤ 5 minutes and all

NFR-10

Maintainability

Database schema migrations — All changes to the database schema shall be managed through versioned Alembic migration scripts, allowing the schema to be reproduced from scratch via migration history.

VERIFICATION CRITERIA:

VERIFICATION: Drop the database, run all Alembic migrations from the beginning on a clean instance. Pass if the resulting schema

NFR-11

Scalability

Containerised deployment — The full application stack (frontend, backend, database) shall be launchable on any host using a single `docker-compose up` command without manual environment configuration beyond a .env file.

VERIFICATION CRITERIA:

VERIFICATION: Clone the repository onto a clean machine, provide only the .env file, run `docker-compose up`. Pass if all services

NFR-12

Data Integrity

Answer persistence — Every answer submitted by a user shall be durably stored in PostgreSQL. No submitted answer shall be lost in the event of an application server restart or crash.

VERIFICATION CRITERIA:

VERIFICATION: Submit 20 answers, simulate an application container restart (docker restart), then query the database. Pass if all

NFR-13

Scoring Accuracy

Scoring consistency — The AI scoring engine shall produce identical scores for identical answer–rubric pairs across repeated invocations, ensuring deterministic output with no randomness in the scoring pipeline.

VERIFICATION CRITERIA:

VERIFICATION: Submit the same answer to the same question 10 times in sequence. Pass if all 10 returned score sets are identical across all four dimensions (Technical Depth, Clarity, Completeness, Structure).

Interview Prep Platform — Requirements Summary

REQUIREMENTS SUMMARY

ID	Category	Requirement (short title)	
FR-01	Authentication	User registration, login & JWT session management	Functional
FR-02	Role Selection	Select SWE / Data Science / PM / Behavioral track	Functional
FR-03	Question Delivery	Retrieve & display role-specific questions one-by-one	Functional
FR-04	Answer Submission	Accept & persist free-text answers with timestamp	Functional
FR-05	AI Scoring	Embed answer; compute cosine similarity vs rubric	Functional
FR-06	Score Display	Show 4-axis score (Depth, Clarity, Completeness, Str)	Functional
FR-07	Keyword Coverage	Identify & surface addressed vs missing rubric concepts	Functional
FR-08	Feedback Generation	Produce actionable improvement suggestions per answer	Functional
FR-09	Progress Tracking	Store history; show per-question score trend over time	Functional
FR-10	Radar Chart	Render dynamic multi-dimensional score radar chart	Functional
FR-11	Question Bank Mgmt	Add/update questions & rubric components	Functional
FR-12	Embedding Cache	Redis-based cache to reduce redundant AI computation	Functional
NFR-01	Performance	AI scoring response ≤ 3 s at 50 concurrent users	Non-Functional
NFR-02	Performance	Cache reduces embedding latency by $\geq 70\%$	Non-Functional
NFR-03	Performance	LCP ≤ 2.5 s on primary pages	Non-Functional
NFR-04	Usability	$\geq 4/5$ new users complete core task without help	Non-Functional
NFR-05	Usability	Responsive layout 375 px – 1440 px, no h-scroll	Non-Functional
NFR-06	Reliability	$\geq 99\%$ uptime over 9-week evaluation period	Non-Functional
NFR-07	Security	JWT auth enforced; 401 on all invalid requests	Non-Functional
NFR-08	Security	Passwords hashed with bcrypt cost factor ≥ 12	Non-Functional
NFR-09	Maintainability	CI pipeline completes in ≤ 5 min on main branch	Non-Functional
NFR-10	Maintainability	Schema fully reproducible via Alembic migrations	Non-Functional
NFR-11	Scalability	Full stack launchable via single docker-compose up	Non-Functional
NFR-12	Data Integrity	Zero answer loss on app restart / container crash	Non-Functional
NFR-13	Scoring Accuracy	Deterministic: identical input always = identical score	Non-Functional

Total: 12 Functional Requirements + 13 Non-Functional Requirements = 25 Requirements